

Mathematics Exam - Test Sheet

Task 1 - Analysis: Limits and Derivatives

Let $f(x) = (x^3 - 8) / (x - 2)$. Examine the following:

- a) Calculate $\lim_{x \rightarrow 2} f(x)$.
- b) Determine $f'(x)$ for $x \neq 2$ (i.e. the derivative of $x^2 + 2x + 4$).
- c) Find all critical points of $g(x) = x^4 - 4x^2$ and classify them as local minima, maxima, or saddle points.

Task 2 - Linear Algebra: Systems and Eigenvalues

Given the matrix A and vector b :

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ 0 & 2 & 3 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ 7 \\ 4 \end{bmatrix}$$

- a) Compute the determinant of A and check whether A is invertible.
- b) Solve the linear system $Ax = b$.
- c) Find the eigenvalues of A and determine whether A is diagonalizable.

Task 3 - Integral Calculus and Orthogonality

Consider the function $h(x) = x \cdot \exp(-x)$ on $[0, \infty)$.

- a) Compute the improper integral: integral from 0 to ∞ of $x \cdot \exp(-x) dx$.
- b) Find the indefinite integral of $h(x) = x \cdot \exp(-x)$ (include constant C).

Given the vectors $v_1 = [1, 2, 2]$ and $v_2 = [2, 1, -2]$:

- c) Check whether v_1 and v_2 are orthogonal.
- d) Compute the projection of v_1 onto v_2 .